∴ number of diffraction lines $= (1 \times 2) + 1 = 3$

16 **B** Distance, $r = 20 \times 10^{16}$ m

For $\lambda_{violet} = 400$ nm

$$s = r\theta = r\frac{\lambda}{D} = \frac{\left(20 \times 10^{16} \times 400 \times 10^{-9}\right)}{3.0} = 2.667 \times 10^{10} = 2.7 \times 10^{10} \text{ m}$$

For $\lambda_{red} = 700$ nm

$$s = r\theta = r\frac{\lambda}{D} = \frac{\left(20 \times 10^{16} \times 700 \times 10^{-9}\right)}{3.0} = 4.667 \times 10^{10} = 4.7 \times 10^{10} \text{ m}$$

For the range of visible light, the order of magnitude of separation ($s = r\theta$) is 10^{10} m.